iPerturb: A Biologically Informed and Explainable Framework for Context-Specific Single-Cell Perturbation Response Modelling

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Abstract. Building an interpretable model that predicts genome-wide transcriptional responses to genetic perturbations remains a grand challenge in biology. Mechanistic kinetic models are difficult to scale to the genome, whereas deep neural network–based models often lack interpretability. To bridge this gap, we propose iPerturb, a biologically informed and explainable framework for context-specific single-cell perturbation response modelling. iPerturb first integrates five tiers of regulatory and association evidence into a cell-line-specific gene regulatory network (GRN) from carefully quality-controlled Perturb-seq data; then propagates a perturbation through it over multiple hops, with a clamped steady state whose convergence we prove and a learnable transfer function that follows the Hill function, aggregated as a matrix product; and finally explains each prediction through the regulatory paths and named neighbours that support it. Evaluated on held-out CRISPRi targets in two human cell lines (702 in K562 and 746 in RPE1), the tiered GRN raises the centred correlation of the predicted response over a linear baseline by +0.093 [+0.077, +0.108] and +0.082 [+0.059, +0.106] (95% intervals), whereas degree-preserving rewired and data-derived graphs give none; the cell-line-specific enhancer tier differs strongly between lines but adds no measurable predictive gain. The matrix-product form lowers peak memory 6.4- to 8.3-fold on a dense graph, and a Chebyshev form of the Hill curve, which reproduces exact Hill propagation to 3.5×10^{-5}, makes edge-specific learnable shapes feasible. On the curated signed core, which reaches 5.8–7.1% of the perturbed genes, the explanation predicts the direction of direct regulation (0.83 and 0.76 sign accuracy at the first hop) but not of indirect effects, and Hill-form propagation does not predict better than linear propagation.

Keywords: gene expression prediction · genetic perturbation · gene regulatory network · biologically informed neural network · Hill function · interpretable AI

1 Introduction

Deciphering how cells transcriptionally respond to perturbations remains a central challenge in biology. Addressing it requires models that are predictive, scal-

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able to the genome, and mechanistically interpretable [3]. Mechanistic kinetic models do not scale, as measuring kinetic parameters experimentally is expensive, limiting them to simple systems [14,12]. Deep neural network–based methods such as GEARS, scGPT, and STATE [19,24,5,15,1] scale to genome-wide prediction but require large training sets and offer little interpretability, and recent benchmarks show that they do not consistently surpass simple linear baselines [2]. Biologically informed neural networks (biNNs) aim to combine both via architectural priors, but existing biNNs mainly predict higher-level phenotypes [10,18] rather than genome-wide expression responses to genetic perturbations.

In this paper, we introduce iPerturb, a biNN-based framework for predicting genome-wide expression changes induced by genetic perturbations, built on two ideas. As illustrated in Figure 1, iPerturb takes a cell line and a genetic perturbation as input and works in three steps. First, cell-line context is used to query public regulatory resources and construct a *context-specific gene regulatory network* (GRN) from five evidence tiers, keeping signed regulation and unsigned association evidence strictly apart. Second, the perturbation is propagated through the GRN by a *learnable transfer function that follows the Hill function of gene regulation* [29,6], with the perturbed gene clamped at every update. Aggregating the messages as a matrix product instead of an edge-wise gather and scatter makes this propagation efficient on accelerators, and expressing the Hill curve of every edge in a Chebyshev basis, a form related to basis-function expansions in learnable-activation networks [17,16], lets even edge-specific learnable shapes run as a single matrix product. Third, every prediction is explained by the named neighbours and edges that support it, and the explanation is tested by ablation.

We applied iPerturb to CRISPRi perturbations in two human cell lines, K562 and RPE1, predicting the mean response of held-out target genes. Our results show that (i) the tiered GRN carries predictive signal for unseen perturbations that degree-preserving rewired and purely data-derived graphs do not, and that this signal survives the removal of response magnitude; the tiers that carry it (curated regulation and protein association) are shared between cell lines, and the cell-line-specific enhancer tier, which differs strongly between lines, adds no measurable predictive gain in K562; (ii) a Chebyshev form reproduces exact Hill propagation to 3.5×10^{-5} while the matrix-product form lowers memory 6.4- to 8.3-fold on a dense graph and makes edge-specific learnable shapes feasible; and (iii) on the curated signed core, which reaches only a small fraction of targets, Hill-form propagation does not predict better than linear propagation. Here, *interpretable* is meant mechanistically: each predicted change decomposes into contributions from named neighbours and edges of the curated network, rather than from opaque latent factors; we do not claim that the explanation paths are validated molecular mechanisms.

2 Related Work

Mechanistic and logic-based models of regulation. Continuous logic models turn regulatory interactions into ordinary differential equations with Hill-type transfer functions [29], and cell-type-specific versions with a modified Hill transfer (TF4) and an edge-strength interpretation have been fitted to signalling data [6]. These models are interpretable but have been applied to networks of tens to hundreds of nodes; iPerturb uses the same transfer function on genome-scale networks, shares its shape parameter per regulator to keep the parameter count manageable, and makes the propagation a matrix product.

Biologically informed neural networks. Genetic Neural Networks arrange one layer per gene so that each gene is predicted from its transcription-factor regulators with Hill-kinetics activations, ordering the layers topologically after removing feedback edges [7]; knowledge-primed and pathway-informed networks predict higher-level phenotypes from curated structure [10,18,8,25]. iPerturb keeps feedback and solves it to a unique convergence point, and it predicts genome-wide expression responses of unseen perturbations rather than a phenotype. Network-based simulation of perturbations has been used to follow transcription-factor knockouts through an inferred GRN by repeated matrix multiplication [13]; we add a sustained clamp, a signed steady state with a proven convergence point, and path-level evidence.

Learned perturbation-response models and their evaluation. Latent-shift and compositional autoencoders [20,19], graph models over gene–gene knowledge [24], and foundation or language models [5,15,1] predict responses to unseen perturbations, but benchmarks show that they often do not surpass simple linear baselines [2] and that systematic shared variation can dominate standard metrics [28]. We therefore evaluate against a linear low-rank baseline, report centred and magnitude-residualised correlations, and use matched degree-preserving controls for every graph.

Network propagation. Propagating labels or scores over biological networks, by random walks with restart, diffusion kernels or truncated walks, amplifies weak signals by integrating all paths between genes and down-weights paths through hubs [4]. iPerturb uses propagation in two ways: signed, clamped propagation of the perturbation itself over the regulatory network (the explanation channel), and propagation of observed training responses over the association network (the predictive channel). Learnable univariate functions in a basis expansion, whose evaluation reduces to matrix multiplication, are used in learnable-activation networks [17,16]; we use the same idea to keep Hill-shaped transfer functions monotone, bounded and cheap, in particular for edge-specific shapes.

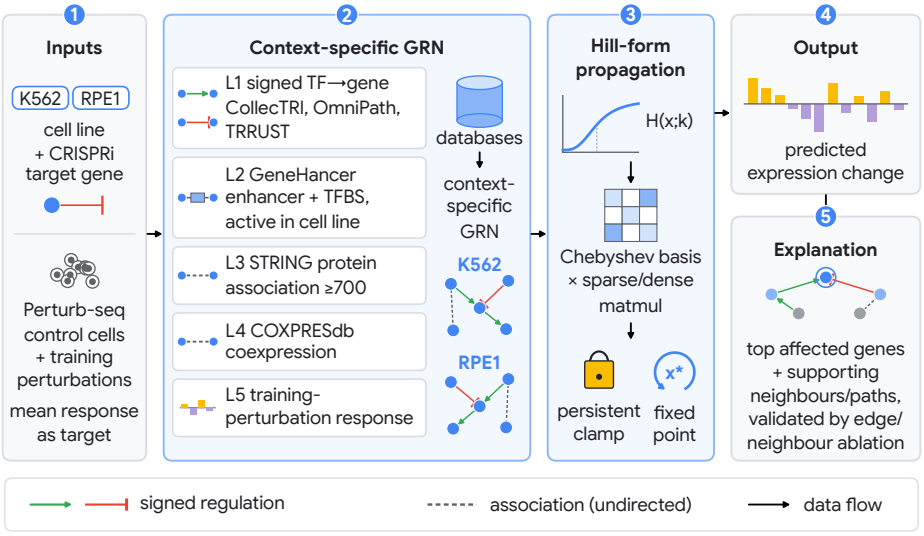


Fig. 1. Overview of the iPerturb framework. (1) A cell line and a CRISPRi target are the inputs; the response is the perturbation-conditioned mean expression change. (2) A context-specific GRN is assembled from five evidence tiers (L1 curated signed regulation; L2 enhancers and binding sites active in the cell line; L3 protein association; L4 co-expression; L5 training-perturbation response), with association evidence kept undirected and unsigned. (3) The perturbation is propagated by a learnable transfer function of Hill form, evaluated in a Chebyshev basis as a matrix product, with the perturbed gene clamped. (4) The predicted expression change is (5) explained by supporting neighbours and edges, tested by ablation. Graphs and bars are illustrative.

3 Data and Methods

3.1 Datasets

Public gene-regulation resources were curated from four complementary source types: (i) transcription factor (TF)–target interactions from TRRUST [11], CollecTRI [21] and OmniPath [27] (120,312 unique directed pairs, 61,215 of them with a defensible sign); (ii) enhancer–gene and TF-binding annotations from GeneHancer v5.26 [9]; (iii) protein–protein association pairs from STRING [26] filtered at combined score ≥ 700; and (iv) co-expression pairs from COXPRESdb [22]. Perturbation data are the genome-scale CRISPRi Perturb-seq screens of Replogle et al. [23] in K562 (chronic myelogenous leukaemia) and RPE1 (retinal pigment epithelium) cells. Starting from the raw UMI counts, each cell was normalised to 10^4 counts and log1p-transformed; the response of a perturbation p is the *mean-level* difference $\Delta_p = \bar{x}_p - \bar{x}_0$ between the mean log-normalised expression of its cells and of the control cells of the same gem groups (batch-matched), computed on the stored gene axis (8,563 genes in K562, 8,749 in RPE1). Perturbations with fewer than 20 cells and targets absent from the gene axis were removed,

leaving 1,835 (K562) and 1,928 (RPE1) targets. The data are single-cell, but the predicted endpoint is the perturbation-conditioned mean response; cell-resolved generation is outside the scope of this work.

3.2 Preprocessing

Perturb-seq data are processed in three representations rather than one matrix: control cells for the baseline state, raw UMI counts for count-level effects, and perturbation effects with uncertainty. Cell quality control is applied per gem group with 5-median-absolute-deviation rules on log UMI and log detected genes (both sides) and on the mitochondrial fraction (upper side only; no universal mitochondrial cutoff, because knockdowns can change metabolism); cells carrying more than one guide target are excluded, and control cells are matched to perturbed cells by gem group. We apply no batch correction, no imputation and no cell-cycle regression, since each can remove or create perturbation-specific signal. QC kept 99.6% of the 310,385 K562 cells and 99.8% of the 247,914 RPE1 cells (1,246 and 501 cells failed the detected-gene rule; controls: 10,670 cells in 48 gem groups for K562, 11,475 in 56 for RPE1). The effect of a perturbation is the difference of mean log-normalised expression to the batch-matched controls (Δ, Section 3.1) with a Welch standard error, a pseudobulk $\log_2$ fold change computed from summed UMIs (library-size normalised, pseudocount), and the knockdown efficiency of the target, $\hat{k}_p = 1 - \bar{x}_{p,\text{target}}/\bar{x}_{\text{control,target}}$ on the linear scale, reported only when the target is detected in controls (≥ 5 CPM); otherwise it is left unknown rather than set to complete knockdown. Cells and guides are not treated as independent replicates: the standard error is descriptive and is used for response labels and precision weights only. Gene sets, graph edges, loadings and all hyper-parameters are fitted on training perturbations only; the 8k-gene axis is the full stored gene axis of the screen (targets are kept even when they are not highly variable).

3.3 Constructing the context-specific GRN

iPerturb integrates evidence into a five-tier hierarchy ordered by strength (Level 1: curated, signed TF–target interactions; Level 2: GeneHancer enhancer–gene links whose enhancers are active in the target cell line and whose TF-binding sites match the TF; Level 3: STRING associations, added as undirected, unsigned pairs; Level 4: COXPRESdb co-expression, top partners per gene; Level 5: signed edges read from training perturbations only). Association evidence (Levels 3–4) is *never* represented as a signed or directed regulatory arrow: it is stored as an undirected edge and used only by the association channel of Section 3.4. Cross-tier duplicates were resolved in favour of the lowest-numbered tier, and within-tier sign conflicts defaulted to unsigned, so an unknown sign is never promoted to activation. The graph is built separately for every cell line: Level 2 depends on the enhancers and TF-binding sites active in that line, and Level 5, when used, is fitted on the training perturbations of that line alone.

3.4 Learnable propagation that follows the Hill function

Hill-form regulatory propagation. Let $x_i \in (\varepsilon, 1-\varepsilon)$ be the regulatory activity of gene i and $x_i^0 = \sigma(z_i)$ its baseline, where z_i is the standardised control expression (a fixed monotone map, so the graph cannot be bypassed by a learned encoder). A regulator j passes its activity through a modified Hill transfer function

$$H(x; k, n) = \frac{k^n \left[1 - (1-x)^n\right]}{(1-x)^n + k^n}, \qquad H(0) = 0,\ H(1) = 1, \tag{1}$$

(the TF4 form of the logic-ODE literature [29,6]) with $n = 2$ and one learnable $k_j \in [0.05, 1]$ *per regulator*, shared by all of its outgoing edges. Signed messages are baseline-centred, so an unperturbed cell produces exactly zero response:

$$x_i^{t+1} = (1-\eta)\, x_i^t + \eta\, \sigma\Big(\operatorname{logit} x_i^0 + \sum_{j \to i} w_{ji} \big[H(x_j^t; k_j) - H(x_j^0; k_j)\big]\Big), \tag{2}$$

with $w_{ji} = \rho\, s_{ji}\, c_{ji}\, \mathrm{softplus}(a) / \max\{1, \sum_{j'} |\cdot|\}$, where $s_{ji} \in \{-1, +1\}$ is the curated sign, c_{ji} the evidence confidence, and ρ, a are learnable scalars. A CRISPRi knockdown of gene p is a *persistent clamp*, $x_p^t = (1-\zeta) x_p^0$ at every iteration, where ζ is the median knockdown of the training perturbations. The fixed point x^* is found by damped iteration ($\eta = 0.5$, 24 steps) and the predicted response is $\hat{\Delta}_g = d_g\,(x_g^* - x_g^0)$ with a non-negative diagonal decoder d_g; no dense $G \times G$ decoder is used, so every prediction decomposes over named edges.

Matrix-multiplication form. Per-edge Hill evaluation requires gathering x_j for every edge and scattering messages, which maps poorly to accelerators. We therefore approximate each regulator's Hill curve by a Chebyshev expansion of degree r,

$$H(x; k_j) \approx \sum_{b=0}^{r} a_b(k_j)\, T_b(2x - 1), \tag{3}$$

whose coefficients $a_b(k)$ are least-squares fitted to Eq. (1) on a logarithmic grid of k and linearly interpolated in $\log k$, so that k_j stays learnable and differentiable. With $\mathbf{h} = \hat{H}(x) - \hat{H}(x^0) \in \mathbb{R}^{B \times G}$ for a batch of B perturbations, the aggregation in Eq. (2) is one sparse (or dense) matrix product $\mathbf{h}\,\mathbf{W}$, with $\mathbf{W}_{ji} = w_{ji}$, instead of an edge-wise gather and index-add. The same fixed-point equations are solved; only the arithmetic changes (equivalence is verified in Section 4.3).

Clamped steady state and multi-hop explanation. For explanations the same network is read as a signed linear system around the baseline. With the perturbed gene g clamped and all other genes R free,

$$\Delta x_R = (I - W_{RR})^{-1} W_{Rg}\, \Delta x_g = \sum_{k \geq 1} W_{RR}^{k-1} W_{Rg}\, \Delta x_g, \tag{4}$$

where W_{ij} is the signed, confidence-weighted effect of gene j on gene i (activation > 0, repression < 0, unknown signs excluded) and Δx_g is the log-expression

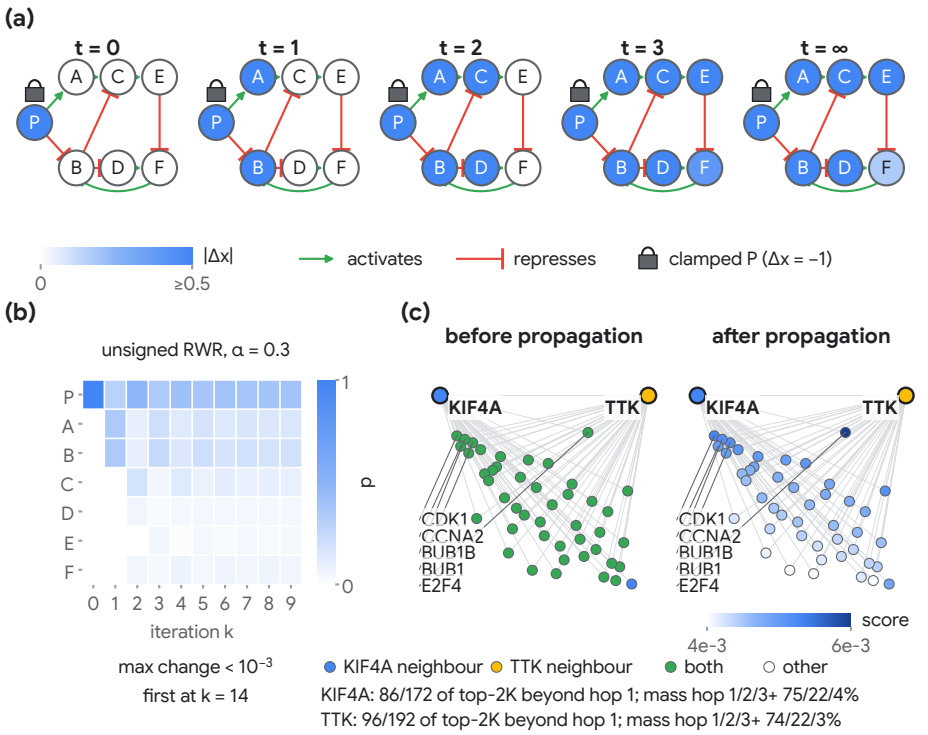


Fig. 2. Multi-hop propagation in iPerturb. **(a)** A CRISPRi perturbation (clamped gene) propagates over a small signed network; node colour is the predicted change after $t = 1, 2, 3$ hops and at the convergence point. **(b)** The association channel propagates signal over unsigned edges by a random walk with restart. **(c)** On the K562 association network, propagation from two genes assigns mass beyond their direct neighbours.

change implied by the partial knockdown (training-median efficiency, not a complete knockout). The k-th term is the contribution of regulatory walks of length k, so every predicted change decomposes into hops and into signed paths; the strongest paths to a gene are reported together with the databases that support each edge and with the reinforcing and opposing path mass, so that cancellation is visible. We scale W so that $\rho(|W|) = 0.9$; Theorem 2 in Appendix E shows that this makes the multi-hop series converge absolutely for every clamped gene (and Theorem 1 gives a unique convergence point for the nonlinear Hill-form iteration, whose gain is parametrised so that it is a contraction for every parameter value). Paths are candidate mechanisms, not proven causal routes.

Response propagation over the association network. The association channel generalises the neighbour mean to multi-hop propagation of *training responses* (network propagation [4]). Seeds are training targets carrying their observed response vectors, and a test gene p receives $\hat{\Delta}_p = \sum_j S_{pj} \Delta_j / \sum_j S_{pj}$ over training targets

j, where S is a propagation kernel of the association network: a random walk with restart, $S = \alpha(I - (1-\alpha)W)^{-1}$ with $W = AD^{-1}$ or $D^{-1/2}AD^{-1/2}$, a heat kernel e^{-tL}, or a walk truncated after K hops, with an optional degree-bias correction (division by $\deg_j^{\gamma}$) and a blend weight with the linear baseline selected on validation perturbations; the kernels converge for every $\alpha \in (0, 1]$ and give convex combinations of training responses (Proposition 3). Networks of different tiers can be propagated separately and combined with non-negative weights fitted on validation perturbations.

Association channel. Most perturbed genes are not curated TFs. Let $\mathbf{G} \in \mathbb{R}^{G \times k}$ be the top-k gene loadings of the centred training responses, and $\hat{A}$ the row-normalised adjacency of the (undirected) context-specific GRN. The embedding of a perturbed gene t is smoothed over its neighbours, $e_t = (1-\alpha)\mathbf{G}_t + \alpha\,(\hat{A}\mathbf{G})_t$, and the response is $\hat{\Delta}_t = \bar{\Delta} + e_t\,\mathbf{W}_e\,\mathbf{G}^{\top}$ with a ridge-fitted $\mathbf{W}_e$; k and α are chosen on the validation perturbations from fixed grids. For $\alpha = 0$ this is the standard linear low-rank baseline, so the channel nests it.

SCC-aware execution. Feedback makes Eq. (2) cyclic. For a frozen graph we compute strongly connected components once, order their condensation topologically, restrict the computation to the nodes reachable from the perturbed gene, settle acyclic singletons with a single forward update and iterate feedback components locally with their upstream inputs fixed. The predictions are those of the global solver (Section 4.3); this is an execution strategy, not a different model.

3.5 Evaluating iPerturb

For each cell line, targets were split into training, validation and test perturbations (70/15/15%) in three folds stratified only by properties known without response data (TF status, control-expression tertile, curated out-degree bin); no target is shared across the splits of a fold. All graph-derived quantities, loadings, knockdown efficacy and hyper-parameters use training (and validation) perturbations only. Because the mean response of a perturbation is dominated by a response shared across perturbations [28], the primary metric is the Pearson correlation of Δ *centred* on the training-mean response (ρ_c), computed per perturbation without the perturbed gene and averaged over perturbations; we also report the same correlation after projecting the training-mean direction out of both profiles (ρ_r, which removes response magnitude), the top-20 centred correlation (ρ_{20}), the mean normalised rank of the correct perturbation among all test perturbations (Cent.; 0.5 is chance) and the area under the precision–recall curve for differentially expressed genes ($|\text{Welch } z| > 4$, cell level, no biological replicates, prevalence-dominated by the shared response). Intervals are 95% bootstrap intervals over unique held-out targets (each target averaged over the folds in which it is tested); paired differences use the same bootstrap. We compare against no change, the training-mean response, a ridge model on a low-rank

gene embedding [2], GEARS [24] trained on the same folds (Section 4.2), a two-layer graph convolutional network on the same GRN, and control GRNs with degree-preserving rewired edges or a graph learned from the training data alone. The full protocol was fixed in a hashed pre-registration file before fitting, and later analyses are labelled as such in the repository.

4 Results

4.1 iPerturb assembles cell-line-specific tiers around a small signed core

Context enters the network through the GeneHancer tier (Level 2) and, when used, through the training-perturbation tier (Level 5); the curated tier (Level 1) and the STRING tier (Level 3) are the same resources for every cell line and differ between lines only through the gene axis. Restricted to enhancers active in each line and to TFs whose binding sites match, the GeneHancer tier contains 645,570 TF–target pairs for K562, 335,734 for HepG2 and 80,414 for Jurkat, yet the edge sets overlap little (Jaccard index 0.08 K562–HepG2, 0.07 K562–Jurkat, 0.05 HepG2–Jurkat), and TF out-degrees agree only moderately across lines (Spearman $\rho = 0.31$ K562–HepG2 to 0.71 K562–Jurkat). The RPE1 tier is empty in this release because no RPE1-labelled TF-binding annotation is available to match, so RPE1 relies on the other tiers. The curated signed core is small relative to the screens. Only 5.8% (K562) and 7.1% (RPE1) of the perturbed genes have a signed out-edge, and 4.4% and 4.7% reach at least 100 genes. Where signed edges exist they are informative: on training perturbations, the sign of a curated edge predicts the direction of the knockdown response of its target in 80% (K562; 109 responsive edges) and 68% (RPE1; 222) of cases (binomial $p < 4 \times 10^{-8}$; edges that share a TF are not independent, and a per-TF test over the 8 and 9 TFs with at least five responsive edges gives $p = 0.008$ and 0.012). The core is therefore reliable but narrow, which motivates the association channel of Section 3.4.

Preprocessing. The careful pipeline of Section 3.2 changes the targets very little: responses recomputed after per-gem-group QC correlate with the earlier ones at a median of 1.0000 (K562; 5th percentile 0.990) and 1.0000 (RPE1; 0.996), as expected for screens that were already filtered. Knockdown is partial and heterogeneous: the knockdown of the target was detectable for 1,835 K562 and 1,927 RPE1 targets, with a median efficiency of 84% and 91%, and 8.1% and 5.0% of targets below 50%, yet efficiency is essentially uncorrelated with the size of the response (Spearman $\rho = 0.03$ and -0.07). Weighting training perturbations and the other design choices that did not matter are examined in Section 5.

4.2 The tiered GRN improves prediction of held-out perturbations

Because the mean response is dominated by a component shared across perturbations, raw correlation is saturated by the training mean (0.26 in K562 and

0.49 in RPE1), and we therefore report centred correlations (Figure 3, Table 1). The smoothing head of iPerturb, in which a perturbed gene's embedding is averaged over its neighbours in the GRN built from the curated and protein-association tiers (L1+L3), improves on the linear low-rank baseline by +0.093 [+0.077, +0.108] (K562) and +0.082 [+0.059, +0.106] (RPE1) in ρ_c, and by +0.083 [+0.069, +0.097] and +0.071 [+0.059, +0.084] after removing the shared response direction (ρ_r), so the gain is not a change in response magnitude. The same model on a degree-preserving rewired GRN gains nothing (the real GRN is better by +0.097 [+0.081, +0.113] and +0.089 [+0.066, +0.113]), and neither does a graph learned from the training data alone with the same degree sequence (the GRN is better by +0.091 [+0.076, +0.107] and +0.080 [+0.057, +0.104]). Zeroing each training row's own-target entry before the decomposition leaves the gain unchanged (+0.098 [+0.084, +0.113], +0.089 [+0.066, +0.113]). The curated regulatory tier adds nothing beyond STRING in this channel (+0.003 [-0.001, +0.007] and +0.002 [-0.006, +0.011] for L1+L3 relative to STRING alone): the signal comes from association evidence, i.e. from the response of functionally related genes, and the case with the largest gain (Section 4.4) is supported by ribosomal-protein neighbours. Because the STRING combined score cannot be split by evidence channel here, we cannot exclude that part of it reflects text-mined knowledge of such screens. Two simple heads on the same GRN do at least as well: a graph convolutional network (B5) and the mean training response of a gene's neighbours that are themselves training targets, which are better than the smoothing head by +0.029 [+0.019, +0.038] and +0.047 [+0.032, +0.063] (GCN) and by +0.047 [+0.035, +0.060] and +0.050 [+0.030, +0.070] (neighbour mean). The GCN loses its gain on the rewired graph (+0.123 [+0.107, +0.139] and +0.118 [+0.094, +0.141] in favour of the real GRN). We therefore regard the three heads as alternative read-outs of the same GRN; the smoothing head is the one fixed before evaluation. GEARS [24], trained on the same folds with its own gene-ontology graph (6 epochs and at most 100 cells per perturbation, below its defaults; 698 K562 and 741 RPE1 targets in its gene-ontology set; centred on its own mean prediction for training targets), reaches $\rho_c = 0.042$ (K562) and 0.163 (RPE1); relative to it, the smoothing head differs by +0.196 [+0.177, +0.215] and +0.090 [+0.059, +0.120]. Propagating the observed responses of training targets over the GRN in several steps, as in network propagation [4], is the strongest read-out: random walks with restart on L1+L3 reach $\rho_c = 0.291$ (K562) and 0.316 (RPE1), and propagating each tier separately and integrating them with non-negative weights fitted on validation targets reaches 0.319 and 0.333, the best ρ_c of all models in Table 1 (Section 5.2). Among the external models, biolord and CPA with gene-ontology descriptions of unseen targets (CPA at a reduced training budget) collapse towards the mean response (ρ_c and centroid rank at chance), as does the zero-shot in silico knockout of scGPT. The released STATE checkpoints, whose pretraining data overlap with these screens and which are therefore not a held-out comparison, are the strongest external rows: the few-shot checkpoint exceeds the best iPerturb read-out in the centroid rank on K562 and in ρ_c on RPE1, and is below it on the remaining metrics

(Table 1). The selected smoothing settings lie at the edge of the fixed grid (fold 0: k=40, α=0.75; fold 1: k=40, α=0.75; fold 2: k=40, α=0.75), so the reported gain may be a lower bound. The area under the precision–recall curve for differentially expressed genes is reported in Table 1 but is prevalence-dominated: the no-change predictor attains the highest value in RPE1.

4.3 The learnable Hill propagation is faithful to the exact Hill form and efficient as a matrix product

Faithfulness. The Chebyshev form approximates the Hill curve to 1.9×10^{-4} at degree 32 and to 2.7×10^{-3} at degree 16 (worst case over $k \in [0.05, 1]$, Figure 4a). Inside the full fixed-point iteration on the curated K562 graph with heterogeneous regulator constants and 200 clamped targets, the largest difference of the equilibrium from exact Hill propagation is 5.2×10^{-4} at degree 8 and 3.5×10^{-5} at degree 16 (decoded response up to 2.4×10^{-5}, against a response scale of 0.087); sparse and dense products agree exactly. The SCC-aware solver reproduces the global solver to 1×10^{-13} (largest feedback component 297 of 8,563 genes in K562, 321 of 8,749 in RPE1) with a median speed-up of 2.3× (K562) and 2.5× (RPE1) in our reference implementation. Restricting the global solver to the genes reachable from the perturbed gene already gives 1.4× and 1.6×; the component ordering adds a further 1.4× and 1.4×.

Efficiency. We separate two effects. First, aggregating messages as a matrix product instead of an edge-wise gather and scatter, which is independent of the transfer function, is slower on the sparse curated graph on an A100 (37 ms vs. 72 ms per step at batch 128, 24 damped updates) but, on the 247k-edge association graph, reduces the step time from 250 ms to 77 ms and the peak memory from 6949 MB to 1078 MB at batch 128 (960 ms to 208 ms and 27689 MB to 3338 MB at batch 512), i.e. 3.3- to 4.6-fold faster with 6.4- to 8.3-fold less memory (Figure 8c). Second, the Chebyshev expansion itself does not speed up a regulator-shared Hill curve with $n = 2$, whose closed form is already cheap (185 ms at degree 16 and batch 128). Its value lies in edge-specific learnable shapes: edge-wise evaluation of per-edge Chebyshev shapes needs activations of size $B \times E \times r$ and runs out of memory on the 40 GB device at batch 128 with $r = 16$ and at batch 512 with $r = 8$, whereas the matrix form stacks a basis of size $B \times G \times r$ and applies one sparse matrix with Er non-zeros, fitting in 5.1 GB and 9.9 GB at batch 128 (6.2-fold less memory at $r = 8$; Figure 8d; outputs agree to 1×10^{-7}). On the CPU, the edge-wise form was the fastest or within a few per cent of the fastest in every shared-curve configuration measured (Appendix B).

Predictive effect. Whether the Hill form improves accuracy over a linear transfer is tested in Section 5.4; it does not, so the operator is reported as a faithful and efficient implementation of the regulatory model rather than as a source of predictive gain.

4.4 Predictions are explained by named neighbours and tested by ablation

For each test target we attribute the prediction to its association neighbours by the change in the predicted shift when a neighbour is removed (no observed data enter the attribution) and test the attribution by ablation against the observed profile. Removing the five top-attributed neighbours lowers the centred correlation more than removing five random neighbours by +0.008 [+0.001, +0.016] (K562, 651 targets) and +0.023 [+0.005, +0.041] (RPE1, 689); for a single neighbour the difference is not distinguishable from zero (0.000 [-0.002, +0.003], +0.004 [-0.002, +0.011]). The attributions are therefore weakly but consistently faithful (Figure 5a). The largest gain in the pre-specified K562 fold is DNTTIP2, whose three most attributed neighbours are the ribosomal proteins RPS4X, RPS18 and RPS6; the smallest gain is a target without neighbours, for which the shrunken embedding performs worse than the linear model, and the first neighbour-free target shows a small gain (Figure 5b–d). In an exploratory analysis, falling back to the linear prediction for neighbour-free targets changes the centred correlation by at most 0.002, because such targets are only 1.6% (K562) and 1.9% (RPE1) of the test set.

4.5 Explanations follow the regulatory mechanism at the first hop, not beyond

We simulated CRISPRi as a persistent clamp on the signed curated network (CollecTRI, OmniPath and TRRUST, extended with signed SIGNOR transcriptional edges) and computed the steady state of Eq. (4) with one fitted gain, then asked whether the explanation agrees with the measured response of held-out targets that have regulatory out-edges (31 in K562 and 47 in RPE1 test perturbations). Among genes that respond ($|z| > 3$), the predicted direction is correct for 0.83 [0.75, 0.89] of the genes at one hop in K562 (114 genes) and for 0.76 [0.71, 0.80] in RPE1 (358 genes), against 0.73 and 0.70 with permuted signs and 0.69 and 0.58 with rewired edges. The controls lie above chance because most responding genes shift in the same direction after a knockdown, so the relevant gain of the real network at the first hop is about 0.83 vs. 0.73 (K562) and 0.76 vs. 0.70 (RPE1). At two hops the accuracy is 0.51 (K562) and 0.60 (RPE1), at three hops 0.51 and 0.52, and for four or more hops 0.55 and 0.49; none of these exceeds the permuted-sign control, and the correlation between predicted and observed responses across all reached genes is 0.02 and 0.01. The signed network therefore explains direct regulation, which is the biology the databases record, but the indirect signed cascade it predicts is not supported by these endpoint data: knockdown responses at distance are dominated by shared stress and growth programmes that the association channel, not the regulatory path, captures. We accordingly present paths beyond the first hop only as candidate mechanisms. The reported case, SSRP1 in K562, acts through MYC on ribosomal-protein genes (all edges activating, supported by CollecTRI and OmniPath), the predicted down-regulation matches three of the five most responsive genes shown,

and the two with the opposite observed sign (RPLP1, UBA52) are the kind of compensatory response that a purely activating path cannot represent.

5 Ablation Study

We ablate the two components of iPerturb, the context-specific GRN (evidence layers, cell-line tier, further databases, propagation kernels) and the learnable Hill propagation (transfer function, certified variant, loss), and the preprocessing. All differences are paired over unique held-out targets with 95% bootstrap intervals (Table 2).

5.1 Evidence layers and controls

Figure 8b summarises the layers of the GRN and the controls of Section 4.2: STRING alone, the curated tier plus STRING, a degree-preserving rewired version, a graph learned from the training data, and the real graph with own-target entries zeroed. Only the real association evidence changes ρ_c relative to no graph; the curated tier adds nothing beyond STRING, and controls with the same capacity and tuning grid gain nothing.

5.2 Evidence tiers and multi-hop propagation

We replaced the smoothing head by propagation of the observed training responses over each tier alone (Table 3) in the manner of network propagation [4]. Association evidence of protein-level tiers carries the signal: STRING alone reaches $\rho_c = 0.298$ (K562) and 0.313 (RPE1), CORUM complexes and BioGRID interactions are weaker but informative, whereas the curated regulatory tier (L1) and co-expression tiers (L4) stay near the linear baseline (0.145 and 0.170). Adding signed SIGNOR edges to L1 changes ρ_c by +0.015 [+0.006, +0.025] (K562) and +0.013 [-0.002, +0.030] (RPE1). CORUM and BioGRID are worse than STRING (-0.035 [-0.051, -0.020] and -0.058 [-0.070, -0.045] in K562), and GTEx co-expression does not improve on COXPRESdb. Reading the same L1+L3 graph beyond the first hop helps: random walks with restart improve on the one-hop neighbour mean by +0.008 [+0.002, +0.015] (K562) and +0.015 [+0.007, +0.024] (RPE1), and truncated two- to three-step walks by +0.017 [+0.011, +0.023] and +0.020 [+0.008, +0.031]. Finally, propagating each tier separately and combining the read-outs with non-negative weights fitted on validation targets gives a further +0.029 [+0.021, +0.036] (K562) and +0.016 [+0.002, +0.030] (RPE1) over L1+L3 with random walks, and +0.174 [+0.155, +0.194] and +0.163 [+0.133, +0.193] over the linear baseline. The weights concentrate on the protein-association tiers (K562: STRING 0.44, CORUM 0.37, BioGRID 0.05, COXPRESdb 0.08, GTEx co-expression 0.01, curated L1 0.00; RPE1: STRING 0.44, CORUM 0.44), so multi-hop propagation adds measurable accuracy, while the curated regulatory tier contributes explanation rather than prediction. The ChIP-Atlas cell-line tier (L2) does not improve over the linear baseline either (Table 3).

5.3 Cell-line-specific tier

Does the cell-line-specific tier carry cell-line-specific predictive signal? We tested this in K562, the only line with a non-empty tier, with the smoothing head of Section 4.2 and a criterion registered before the run: the K562 tier must beat both foreign-line tiers. It does not. Adding the K562 GeneHancer tier to the L1+L3 graph changes the centred correlation by -0.011 [-0.019, -0.004] relative to the base graph, by -0.005 [-0.011, 0.000] relative to adding the HepG2 tier and by -0.008 [-0.015, 0.000] relative to the Jurkat tier; the enlarged graph is better than its degree-preserving rewired version, which also scrambles L1+L3, by +0.064 [+0.052, +0.076]. The cell-line-specific tier therefore differs strongly between lines but adds no measurable predictive gain over generic association evidence in this setting.

5.4 Hill versus linear propagation

Following the pre-registered gate, we trained the shared-Hill model and a linear model with the same graph, decoder and budget as a residual on the linear baseline for the reachable targets (37 K562 and 50 RPE1 unique targets over three folds). All differences in the centred correlation are below 0.002 in either direction. The Hill form relative to the linear form is +0.0014 [-0.0001, +0.0041] in K562 and +0.0006 [0.0000, +0.0018] in RPE1 (marginally positive in RPE1), relative to the linear baseline without graph +0.0001 [-0.0003, +0.0007] and -0.0004 [-0.0015, +0.0002], and the real graph relative to a rewired graph is +0.0001 [-0.0002, +0.0005] and -0.0004 [-0.0015, +0.0002]; the pre-registered criterion (an advantage over both the linear form and the no-graph baseline, and over the rewired graph) was not met. An oracle readout with its scale fitted on the test targets themselves added nothing, so the limit is the information in the signed core rather than the optimiser. The conclusion is unchanged for a certified, anti-vanishing variant (log-odds state so that large responses are not squashed, a leaky Hill transfer that keeps a gradient where the curve is flat, a contraction guaranteed for every parameter value, Appendix E) trained with a correlation-based loss that does not shrink with the size of the response: relative to no graph, the Hill form changes ρ_c on reachable targets by +0.0001 [-0.0016, +0.0020] (MSE) and -0.0024 [-0.0086, +0.0016] (correlation loss) in K562 and by -0.0003 [-0.0014, +0.0006] and -0.0006 [-0.0026, +0.0010] in RPE1 (Table 5); every interval includes zero. The Hill-form operator is therefore a faithful and efficient implementation of the regulatory model, not a source of additional predictive accuracy on this data.

5.5 Preprocessing

Weighting training perturbations by cell number or by response precision did not help consistently (K562 -0.005 [-0.008, -0.003] and -0.010 [-0.014, -0.006] in ρ_c; RPE1 +0.021 [-0.002, +0.043] and +0.005 [-0.022, +0.033]), so the headline models are unweighted.

6 Conclusion and Discussion

We introduced iPerturb, which builds a cell-line-specific regulatory network from five evidence tiers and propagates perturbations through it with a learnable transfer function that follows the Hill function, so that every prediction can be traced to named genes and edges. On held-out CRISPRi targets in K562 and RPE1 the tiered network carries predictive signal that matched rewired and data-derived graphs do not, multi-hop propagation of training responses over the tiers, integrated across tiers, is the strongest read-out (0.319 and 0.333 in ρ_c against 0.145 and 0.170 for the linear baseline), and the Hill-form operator reproduces exact Hill propagation to 3.5×10^{-5} while its matrix-product form is memory-lean on dense graphs. Two limitations are central. First, the tiers that carry the predictive gain (curated regulation and protein association) are the same for every cell line; the cell-line-specific enhancer tier, testable only in K562, differs strongly between lines but did not improve prediction over a foreign line's tier or over the base graph, so its context specificity is descriptive rather than predictive in this setting. Second, on the curated signed core, which reaches only 5.8–7.1% of the perturbed genes in these essential-gene screens, a nonlinear Hill transfer does not predict better than a linear one, and the gain over a linear low-rank model comes from association evidence rather than from curated regulation. The mechanism-level test sharpens this: the signed network predicts the direction of direct regulatory effects, but the signed multi-hop cascade is not supported beyond the first hop in these endpoint data, so we present longer paths only as candidate mechanisms, and we prove only that the multi-hop quantity is well defined and has a unique convergence point (Appendix E), not that it is correct. We read the coverage limitation as a statement about the screens, not about biology: the screens perturb mostly non-TF genes, and a signed core cannot propagate what it does not reach.

Several limitations qualify these results. The predicted endpoint is the mean response, not a distribution of cells. RNA abundance is a proxy for regulator activity and endpoint data cannot identify time constants, so the fixed-point operator is a steady-state model; neighbour attributions are tested by ablation but are association paths, not validated mechanisms. The folds and seeds reuse the same biological screens, and graph and no-graph models were tuned on comparable but not identical budgets; matched controls share their grid. The largest gains are obtained for genes in well-characterised complexes (for example, ribosomal-protein neighbours), which are enriched in essential-gene screens, and the transfer to screens with few such complexes is untested. The GeneHancer tier is empty for RPE1, so the cell-line-specific tier is tested in K562 only; a test with the training-perturbation tier (Level 5) is left to future work. Future work will add more cell lines and a genome-wide screen with more transcription-factor targets, and test edge-specific learnable shapes, which the matrix-multiplication form now makes affordable, on data in which a signed core has more reach.

Conflict of interests, acknowledgments, funding, and data availability

Conflict of Interests. The authors declare that they have no competing interests.

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Data and Software Availability. The CRISPRi Perturb-seq datasets analysed in this study are publicly available from Replogle et al. [23]. Prior biological knowledge was obtained from TRRUST, CollecTRI, OmniPath, GeneHancer v5.26, STRING v12 and COXPRESdb; GeneHancer is licence-gated and is not redistributed. The source code, the frozen pre-registration of the evaluation protocol with its later addenda, figure-generation scripts and the files from which every number in this paper is generated accompany the submission and will be released in the public iPerturb repository (https://github.com/kagtgi/iPerturb) at publication.

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A Additional protocol details

Splits and fitting. Three target-held-out folds per line (70/15/15%, stratified by TF status, control-expression tertile and curated out-degree bin; seed 42, fold-specific seeds). The smoothing head selects $k \in \{5, 10, 20, 40\}$ and $\alpha \in \{0, 0.25, 0.5, 0.75\}$ on validation perturbations (selected settings: K562, fold 0: k=40, α=0.75; fold 1: k=40, α=0.75; fold 2: k=40, α=0.75; RPE1, fold 0: k=40, α=0.75; fold 1: k=40, α=0.75; fold 2: k=40, α=0.75); the ridge penalty is 10^{-2}. The GCN has two layers, hidden size 64, residual self-weights, AdamW (learning rate 10^{-3} or 3×10^{-3}, weight decay 10^{-3}), at most 200 epochs, patience 20, three seeds. Hill-form models use $n = 2$, damping 0.5, 24 updates, learning rates $\{0.003, 0.01\}$ and penalties $\{10^{-3}, 10^{-1}\}$ on the per-gene decoder offset, selected on validation reachable targets; the clamp strength ζ is the median training knockdown.

Pre-registration. The evaluation protocol was written and hashed before any graph model was fitted. Later analyses (association smoothing, controls, GCN, explanations, context-specific tier, efficiency study, neighbour-free fallback) were added as dated addenda before they were run; the fallback analysis is exploratory and is not part of the headline numbers.

GEARS. Trained with the official package on at most 100 cells per perturbation and 3,000 control cells (log1p of counts per 10^4), for 6 epochs (fewer than the package default, which exceeded the memory of our 83 GB host at the full cell count), hidden size 64, with its default gene-ontology graph; targets absent from its ontology set are excluded from its evaluation. Its predictions are expressed relative to its own control mean, whereas the observed responses use batch-matched controls, so GEARS is centred on the mean of its predictions for a sample of 300 training targets before scoring (the raw correlation is reported in the repository). These settings make GEARS a budget-limited baseline.

B Efficiency measurements

On the CPU, the edge-wise form was the fastest for the sparse curated graph at every batch size (387 ms at batch 128 against 1485 ms for the sparse product), and on the association graph the dense product matched it (3477 ms vs. 3580 ms).

C External baselines

All external models were scored with the folds, metrics and centring of the other models (predictions are centred on the mean prediction for training targets). *STATE*: the released State Transition checkpoints for Replogle-HVG data (zero-shot and few-shot variants, K562 and RPE1), evaluated without retraining; their pretraining data overlap with these screens (marked †). *scGPT*: the whole-human checkpoint without fine-tuning; the official perturbation head is trained from scratch, so we ran a zero-shot in silico knockout (target expression set to zero in 100 control cells, change of the predicted expression averaged over cells) on the 1,199 most expressed in-vocabulary genes; targets outside the vocabulary or with near-zero control expression are excluded. *biolord*: official package, with the gene-ontology neighbour vector of the GEARS gene-ontology graph as the attribute of the perturbed gene; no tuning. *CPA*: `cpa-tools` with the same neighbour vectors as frozen perturbation embeddings, the hyper-parameters of its Norman tutorial, at most 50 cells per training target and a wall-clock cap of 25 minutes per fold (a reduced budget).

D Data replay

An independent re-computation of the target of ten random K562 perturbations directly from the raw counts (no chunking or sparse accumulation) reproduced the stored responses to within 4.5×10^{-8} (single-precision rounding); the script and its output are in the repository.

E Convergence of multi-hop propagation

We show that the multi-hop propagation used by iPerturb has a unique convergence point (a fixed point) that is reached geometrically from any starting state, and that the hop-by-hop decomposition used for explanations is well defined. Throughout, $W \in \mathbb{R}^{G\times G}$ holds the signed effect of gene j on gene i (W_{ij}), g is the clamped (perturbed) gene, R the set of free genes, $\|W\|_\infty = \max_i \sum_j |W_{ij}|$ is the largest total incoming weight, $\rho(\cdot)$ the spectral radius and $|W|$ the entry-wise absolute value. Convergence is a property of the operator; it does not make a prediction correct (Remark 2).

Operators. (O1) *linear:* $\Delta x_R = W_{RR}\Delta x_R + W_{Rg}\Delta x_g$. (O2/O3) *Hill-form:* with a state z_R ($z = \operatorname{logit} x$; the sigmoid-state variant iterates x directly), baseline z^0 and clamped z_g,

$$F_i(z) = z_i^0 + \sum_j W_{ij}\big[T(\sigma(z_j)) - T(\sigma(z_j^0))\big], \qquad T(x) = (1-\lambda)H(x;k_j) + \lambda x, \quad (5)$$

iterated as $z^{t+1} = (1-\eta)z^t + \eta F(z^t)$ with the clamped coordinate held fixed at every step.

Lemma 1 (Hill slope). *For $n = 2$ and $k \le \sqrt{3}$, $L_H(k) := \max_{x\in[0,1]} |H'(x;k)| = \dfrac{9(1+k^2)}{8\sqrt{3}\,k}$, attained at $1 - x = k/\sqrt{3}$. With leak λ, $L_T \le (1-\lambda)L_H(k) + \lambda$.*

Proof. With $u = 1-x$, $H = k^2(1-u^2)/(u^2+k^2)$ and $|H'| = 2u\,k^2(1+k^2)/(u^2+k^2)^2$. The derivative of $u/(u^2+k^2)^2$ has the sign of $k^2 - 3u^2$, so the maximum is at $u = k/\sqrt{3} \le 1$, where $u^2 + k^2 = 4k^2/3$ and $|H'| = 9(1+k^2)/(8\sqrt{3}k)$. The leak adds slope λ and removes at most λL_H.

Numerically $L_H = 13.02$ at $k = 0.05$, 2.36 at $k = 0.3$ and 1.30 at $k = 1$; the formula agrees with a dense-grid maximum in the test suite.

Proposition 1 (Existence). *For every W, every parameter value and every clamp, the map (5) has a fixed point.*

Proof. T takes values in $[0,1]$, so each bracket in (5) lies in $[-1,1]$ and F maps $\mathbb{R}^R$ into the compact convex box $\{|z_i - z_i^0| \le \|W\|_\infty\}$; F is continuous, and Brouwer's theorem applies. (For the sigmoid state the box is $[0,1]^R$.)

Theorem 1 (Uniqueness and geometric convergence). *Let $q := \frac{L_T}{4}\|W_{RR}\|_\infty < 1$ and $q_\eta := 1 - \eta(1-q)$. Then the damped clamped iteration is a contraction of $(\mathbb{R}^R, \|\cdot\|_\infty)$ with constant q_η. It has a unique fixed point z^*, which is also the unique fixed point of F, and for every start z^0 and all t,*

$$\|z^t - z^*\|_\infty \le q_\eta^t \|z^0 - z^*\|_\infty, \qquad \|z^t - z^*\|_\infty \le \frac{q_\eta}{1-q_\eta}\|z^t - z^{t-1}\|_\infty.$$

A tolerance τ is reached after $t \ge \ln\big(\tau(1-q_\eta)/\|z^1 - z^0\|_\infty\big)/\ln q_\eta$ steps.

Proof. Since $\sigma' \le 1/4$ and $|T'| \le L_T$, $|F_i(z) - F_i(z')| \le \sum_j |W_{ij}|\, L_T \frac{1}{4}|z_j - z'_j|$, hence $\|F(z) - F(z')\|_\infty \le q\|z - z'\|_\infty$ on the free coordinates (the clamped coordinate does not move). The damped map is the convex combination $(1-\eta)\mathrm{id} + \eta F$, with constant $1 - \eta + \eta q = q_\eta$. Banach's fixed-point theorem gives existence, uniqueness and the first bound; the second follows from the triangle inequality $\|z^t - z^*\| \le \|z^t - z^{t+1}\| + q_\eta\|z^t - z^*\|$ and the bound on $\|z^{t+1} - z^t\| \le q_\eta\|z^t - z^{t-1}\|$.

Corollary 1 (Certified parametrisation). *GraphNet2 sets $\|W\|_\infty = \gamma\,\sigma(\varrho)\,4/L_T(k)$ with $\gamma \le \frac{1}{2}$ (incoming weights normalised to sum one, then scaled by a learnable gain through $\sigma(\varrho)$). Since $\|W_{RR}\|_\infty \le \|W\|_\infty$, $q \le \gamma \le \frac{1}{2}$ for every value of $k \in [0.05, 1]$ and of the learnable parameters, so Theorem 1 holds for every model the optimiser can reach. With $\eta = \frac{1}{2}$, $q_\eta \le \frac{3}{4}$.*

Theorem 2 (Multi-hop response is an absolutely convergent walk sum). *Assume $\rho(|W|) < 1$ (enforced by the scaling in the construction of W). Then for every clamped gene g and free set R: (i) $\rho(|W_{RR}|) \le \rho(|W|) < 1$, so $I - W_{RR}$ is invertible; (ii) $\sum_{k\ge 0} W_{RR}^k W_{Rg} \Delta x_g$ converges absolutely to $(I - W_{RR})^{-1} W_{Rg} \Delta x_g$, so the hop contributions $\Delta x^{(k)} = W_{RR}^{k-1} W_{Rg} \Delta x_g$ and the contributions of individual walks are well defined and may be summed in any order; (iii) with $v = (I - |W|)^{-1}\mathbf{1} \ge \mathbf{1}$ and $q_v := 1 - 1/\max_i v_i < 1$, truncating after K hops leaves an error $\| \sum_{k>K} \Delta x^{(k)} \|_v \le \frac{q_v^K}{1-q_v} \|W_{Rg} \Delta x_g\|_v$, where $\|x\|_v = \max_i |x_i|/v_i$.*

Proof. (i) $|W_{RR}|$ is a principal block of $|W|$; for non-negative matrices the spectral radius is monotone under entry-wise domination and under taking principal blocks (Perron–Frobenius), and $\rho(W_{RR}) \le \rho(|W_{RR}|)$. (ii) $|W_{RR}^k b| \le |W_{RR}|^k |b|$ and $\sum_k |W_{RR}|^k$ converges when $\rho < 1$; absolute convergence permits rearrangement (Fubini) into hops and into walks. (iii) $(I - |W|)^{-1} = \sum_k |W|^k \ge I$ entry-wise, so $v \ge \mathbf{1}$ and $|W|v = v - \mathbf{1} \le (1 - 1/\max v)\, v$. For any x, $|W_{RR}x| \le |W_{RR}||x| \le \|x\|_v |W| v \le q_v \|x\|_v v$ on R, so $\|W_{RR}x\|_v \le q_v \|x\|_v$ and the tail of the geometric series follows.

Remark 1 (Why $\rho(|W|)$). The signed spectral radius does not suffice: $W = \left(\begin{smallmatrix} 1 & 1 \\ -1 & -1 \end{smallmatrix}\right)$ has $\rho(W) = 0$, but the block $W_{RR} = (1)$ for $R = \{1\}$ has $\rho = 1$ and $I - W_{RR}$ is singular. Nor is $\|W\|_\infty < 1$ needed: on the curated graphs the total incoming weight is $\|W\|_\infty = 5.3$ (K562) and 8.0 (RPE1) after scaling to $\rho(|W|) = 0.90$, while $q_v = 0.977$ and 0.983.

Proposition 2 (SCC-aware execution). *Under Theorem 1, iterating the strongly connected components of the free graph in topological order (a single update for acyclic singletons, a local damped iteration for feedback components with already settled inputs) converges to the same unique fixed point as the global iteration; genes not reachable from g stay at their baseline.*

Proof. Each block map is a restriction of F with the constant of a principal block ($\le q_\eta$), hence a contraction for fixed inputs. The block solutions, composed in topological order, satisfy every equation of the full system, and the fixed point is unique, so the two coincide. A gene without a path from g has all parents either unreachable or at baseline, so x_i^0 solves its equation, and by uniqueness equals the fixed point.

Proposition 3 (Response-propagation kernels). *For the association graph let $W = AD^{-1}$ (column-stochastic) and $S_\alpha = \alpha(I - (1-\alpha)W)^{-1}$, $\alpha \in (0, 1]$. The series $\sum_k ((1-\alpha)W)^k$ converges geometrically with ratio $1 - \alpha$, every column of S_α is a probability vector (non-negative, sum one), and the symmetric normalisation $D^{-1/2} A D^{-1/2}$ has spectrum in $[-1, 1]$, so $\rho((1-\alpha)W_{\mathrm{sym}}) \le 1 - \alpha$. The heat kernel e^{-tL}, $L = D - A$, exists for all $t \ge 0$. The propagated prediction $\sum_j S_{pj} \Delta_j / \sum_j S_{pj}$ is a convex combination of training responses and therefore stays in their convex hull.*

Proof. $\|(1-\alpha)W\|_1 = 1-\alpha < 1$ because the columns of W sum to one; the column sums of S_α are $\alpha\sum_k(1-\alpha)^k = 1$; non-negativity follows from the series. L is positive semi-definite, so e^{-tL} is a contraction semigroup.

Remark 2 (What is not claimed). Convergence is not correctness. Endpoint data do not identify dynamics, the fixed point is a steady-state description, and the multi-hop signs beyond one hop are at chance on held-out targets (Section 4.5); the theorems guarantee that the quantity we explain is well defined, not that it matches biology.

The restart-walk kernels have column sums equal to one to within 2×10^{-14} for $\alpha \in \{0.1, 0.5, 0.9\}$.

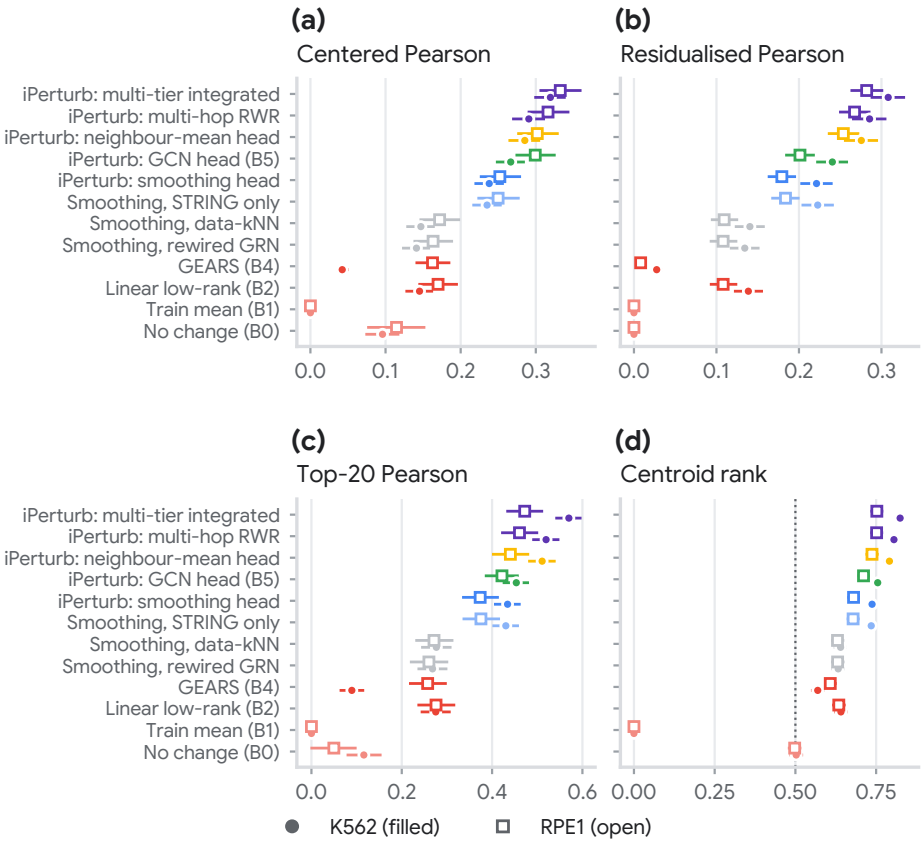


Fig. 3. The tiered GRN improves response prediction for unseen CRISPRi targets in two human cell lines. Mean over held-out targets with 95% bootstrap intervals over unique targets (three target-held-out folds; K562 $n = 702$, RPE1 $n = 746$). Filled circles, K562; open squares, RPE1. **(a)** Pearson correlation of the response after removing the training-mean response (ρ_c, primary; the train mean scores 0 by construction). **(b)** The same after projecting the training-mean direction out of both profiles (ρ_r). **(c)** Top-20 centred correlation. **(d)** Mean normalised rank of the correct perturbation (centroid accuracy; 0.5 is chance). Smoothing with a rewired GRN or a data-kNN graph are controls with the same capacity and tuning grid.

Table 1. Test results on held-out CRISPRi targets. Unique-target means over three folds (bold: best per column). ρ_c: centred correlation; ρ_r: after removing the training-mean direction; ρ_{20}: top-20 centred correlation; Cent.: mean normalised centroid rank; AUPRC: differentially expressed genes (|Welch z| > 4). Intervals for all differences are given in Table 2. †Released pretrained checkpoint evaluated without retraining; its pretraining data overlap with these screens, so the row is not a held-out comparison and is excluded from the bold marking. CPA (B7) and biolord (B8) use the official packages with fixed gene-ontology embeddings for unseen targets (Appendix C). External models are scored on the targets they support.

K562 (702 held-out targets) Model	ρ_c	ρ_r	ρ_{20}	Cent.	AUPRC
No change (B0)	0.096	0.000	0.116	0.503	0.099
Train mean (B1)	0.000	0.000	0.000	0.000	0.031
Linear low-rank (B2)	0.145	0.139	0.275	0.641	0.110
GEARS (B4)	0.042	0.028	0.089	0.569	0.056
CPA (B7)	-0.001	-0.002	-0.004	0.488	0.041
biolord (B8)	0.001	0.005	0.004	0.497	0.059
scGPT, zero-shot knockout†	0.020	-0.004	-0.015	0.511	0.061
STATE, zero-shot†	0.157	0.105	0.347	0.818	0.070
STATE, few-shot†	0.253	0.185	0.472	0.882	0.100
Smoothing, rewired GRN	0.141	0.134	0.268	0.633	0.110
Smoothing, data-kNN graph	0.147	0.141	0.277	0.640	0.112
Smoothing, STRING only	0.235	0.223	0.430	0.735	0.132
iPerturb: smoothing head	0.238	0.221	0.434	0.737	0.132
iPerturb: GCN head (B5)	0.267	0.240	0.454	0.755	0.140
iPerturb: neighbour-mean head	0.285	0.276	0.511	0.791	0.162
iPerturb: multi-hop propagation (RWR)	0.291	0.286	0.520	0.805	0.169
iPerturb: multi-tier integrated propagation	**0.319**	**0.308**	**0.570**	**0.824**	**0.180**
RPE1 (747 held-out targets) Model	ρ_c	ρ_r	ρ_{20}	Cent.	AUPRC
No change (B0)	0.115	0.000	0.050	0.498	**0.245**
Train mean (B1)	0.000	0.000	0.000	0.000	0.054
Linear low-rank (B2)	0.170	0.108	0.276	0.634	0.175
GEARS (B4)	0.163	0.008	0.257	0.608	0.151
CPA (B7)	0.002	0.000	0.012	0.514	0.066
biolord (B8)	-0.000	0.002	-0.004	0.503	0.079
scGPT, zero-shot knockout†	0.030	0.006	-0.053	0.510	0.097
STATE, zero-shot†	0.186	0.088	0.372	0.751	0.115
STATE, few-shot†	0.343	0.151	0.528	0.816	0.143
Smoothing, rewired GRN	0.163	0.108	0.260	0.631	0.174
Smoothing, data-kNN graph	0.172	0.109	0.271	0.630	0.175
Smoothing, STRING only	0.250	0.184	0.375	0.679	0.192
iPerturb: smoothing head	0.252	0.179	0.374	0.680	0.191
iPerturb: GCN head (B5)	0.299	0.201	0.422	0.711	0.191
iPerturb: neighbour-mean head	0.302	0.254	0.440	0.737	0.204
iPerturb: multi-hop propagation (RWR)	0.316	0.267	0.461	0.752	0.205
iPerturb: multi-tier integrated propagation	**0.333**	**0.282**	**0.472**	**0.752**	0.210

Table 2. Paired differences in ρ_c. Mean difference with 95% bootstrap interval over unique held-out targets.

Comparison (ρ_c, mean difference [95% CI])	K562	RPE1
GRN vs no graph	+0.096 [+0.082, +0.110]	+0.087 [+0.064, +0.110]
GRN vs rewired GRN	+0.097 [+0.081, +0.113]	+0.089 [+0.066, +0.113]
GRN vs data-kNN graph	+0.091 [+0.076, +0.107]	+0.080 [+0.057, +0.104]
GRN vs rewired (self-entries zeroed)	+0.098 [+0.084, +0.113]	+0.089 [+0.066, +0.113]
iPerturb vs linear (B2)	+0.093 [+0.077, +0.108]	+0.082 [+0.059, +0.106]
iPerturb vs GEARS	+0.196 [+0.177, +0.215]	+0.090 [+0.059, +0.120]

Table 3. Each evidence tier as a propagation graph, and multi-hop readouts. Unique-target mean ρ_c over three target-held-out folds; the response of a held-out target is the propagated, normalised response of training targets over the graph, with the restart probability (or the number of hops) chosen on validation targets.

Graph (random walk with restart, α selected)	K562 ρ_c	RPE1 ρ_c
Linear low-rank (B2)	0.145	0.170
L1 curated (CollecTRI/OmniPath/TRRUST)	0.131	0.162
L1 + SIGNOR	0.146	0.175
L3 STRING	0.298	0.313
L3 CORUM complexes	0.263	0.278
L3 BioGRID physical	0.241	0.264
L4 COXPRESdb	0.173	0.208
L4 GTEx co-expression	0.170	0.195
L2 ChIP-Atlas (cell line)	0.130	–
L1+L3, one hop (neighbour mean)	0.282	0.301
L1+L3, random walk with restart (multi-hop)	0.291	0.316
L1+L3, truncated 2–3 hops	0.299	0.321
All tiers, propagated separately and integrated	0.319	0.333

Table 4. Time per forward and backward step (ms) and peak memory (MB) of the regulator-shared Hill operator (24 updates, A100 40 GB). E0: exact Hill, edge gather/scatter; E1: exact Hill, sparse matrix product; E2: dense product; C1: degree-16 Chebyshev form, sparse product.

Graph	Batch	E0	E1	E2	C1
curated, 10k edges	128	37 / 942	72 / 1054	104 / 1592	179 / 4001
curated, 10k edges	512	73 / 3708	151 / 3315	346 / 3791	305 / 15018
association, 247k edges	128	250 / 6949	77 / 1078	103 / 1606	185 / 4025
association, 247k edges	512	960 / 27689	208 / 3338	346 / 3803	360 / 15041

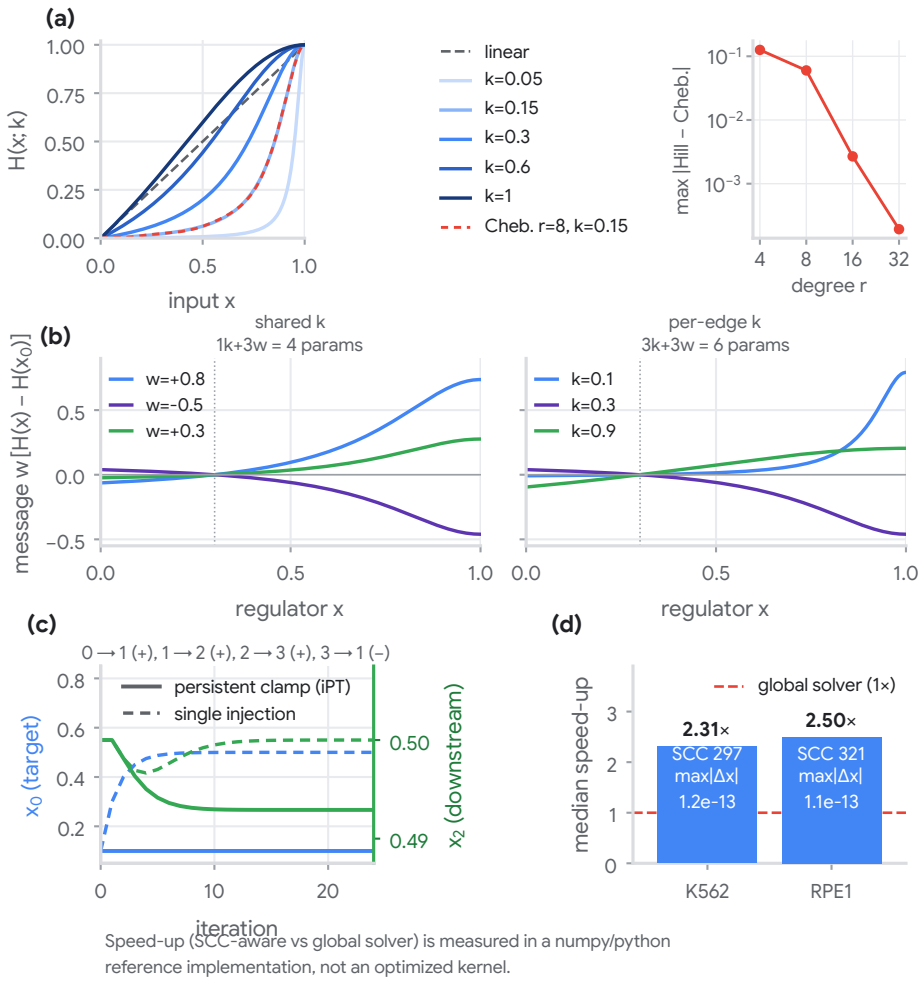
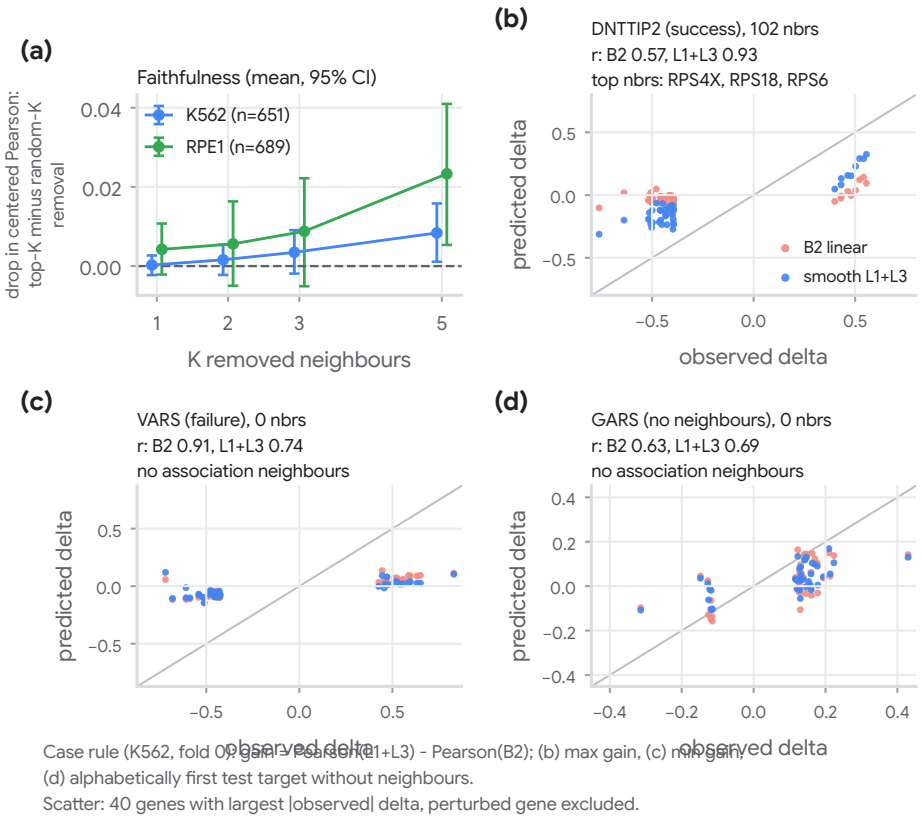


Fig. 4. The regulatory operator. (a) Modified Hill transfer functions of Eq. (1) for five dissociation constants, the linear transfer, and the degree-8 Chebyshev form for $k = 0.15$; the inset shows the maximal approximation error against the degree. **(b)** Messages of one regulator to three targets with a shared k (rescaled copies of one shape) or edge-specific k (three shapes). **(c)** A persistent clamp re-imposes the perturbed gene at every update, whereas a single injection lets it relax (four-gene motif with feedback; x_2 on the right axis). **(d)** Median speed-up of the SCC-aware solver over the global solver (numpy reference implementation).



Case rule (K562, fold 0): gain = Pearson(L1+L3) - Pearson(B2); (b) max gain, (c) min gain,
(d) alphabetically first test target without neighbours.
Scatter: 40 genes with largest |observed| delta, perturbed gene excluded.

Fig. 5. Explanations of the association channel. (a) Removing the top-K attributed neighbours of a test target lowers its agreement with the observed profile more than removing K random neighbours (mean difference with 95% interval; targets with at least six neighbours). **(b–d)** Cases chosen by a rule fixed beforehand (K562 fold 0): largest gain over the linear baseline, smallest gain, and the first target without neighbours. Neighbours are association evidence, not causal paths.

Table 5. Certified, anti-vanishing Hill propagation on reachable held-out targets. Unique-target mean ρ_c and paired difference to the no-graph linear baseline (95% bootstrap interval); all variants use the same graph, decoder, optimiser and tuning grid. Predictions are not numerically collapsed (standard deviation of the learned correction 10^{-3} against a residual standard deviation of about 0.06–0.09), but they carry no usable signal.

Reachable targets	K562 ρ_c	Δ vs B2	RPE1 ρ_c	Δ vs B2
No graph (B2)	0.073	ref.	0.147	ref.
Linear propagation, MSE	0.069	-0.0041 [-0.0139, +0.0017]	0.146	-0.0012 [-0.0045, +0.0010]
Linear propagation, correlation loss	0.067	-0.0055 [-0.0161, +0.0018]	0.145	-0.0022 [-0.0051, -0.0001]
Certified Hill, MSE	0.073	+0.0001 [-0.0016, +0.0020]	0.147	-0.0003 [-0.0014, +0.0006]
Certified Hill, correlation loss	0.070	-0.0024 [-0.0086, +0.0016]	0.146	-0.0006 [-0.0026, +0.0010]
Certified Hill, correlation + Huber	0.070	-0.0024 [-0.0087, +0.0016]	0.146	-0.0006 [-0.0026, +0.0010]

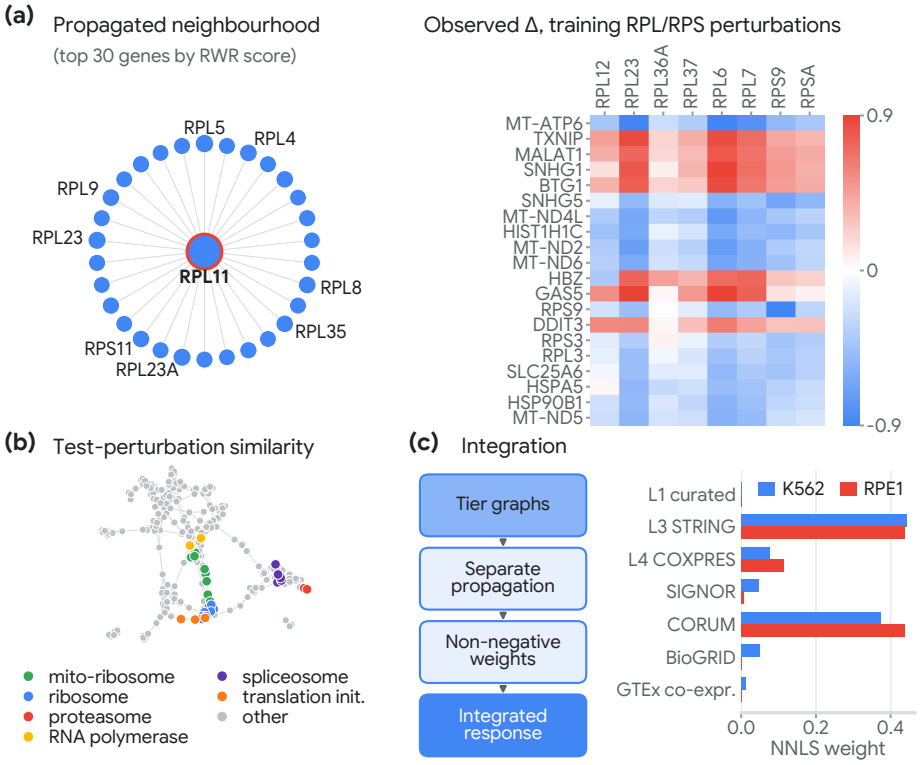


Fig. 6. Propagation over the association network recovers functional modules. (a) Random-walk-with-restart neighbourhood of the held-out ribosomal gene RPL11 (K562, fold 0; node area proportional to propagation score; all 30 top-scoring genes are ribosomal proteins) and the observed responses of eight training ribosomal perturbations, whose profiles correlate at 0.52 on average. **(b)** Test perturbations joined to their three most similar predicted profiles; edges between family-annotated nodes join the same family in 86.0% of 43 cases against 21.1% expected by chance. **(c)** Integration of tier-specific propagations with non-negative weights (mean over three folds); the protein-association tiers carry most of the weight, the curated tier none.

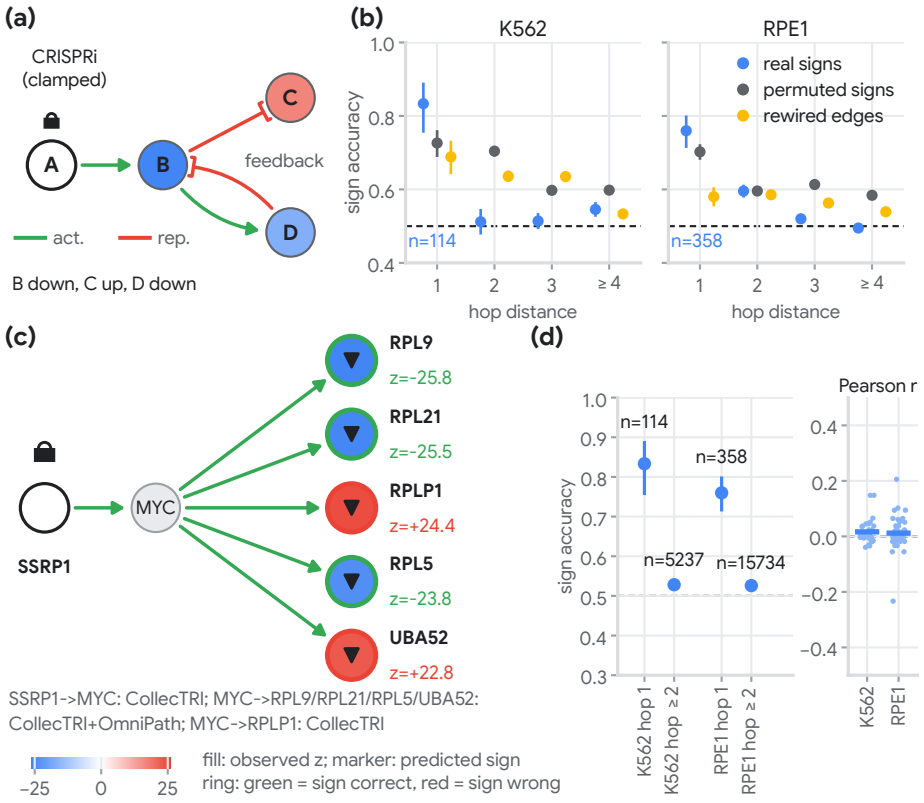


Fig. 7. Mechanism-level explanation and its test. (a) A CRISPRi perturbation of A clamps A and lowers B, raises C (repressed by B) and lowers D, as computed by the model. **(b)** Sign accuracy of the predicted direction among responding genes ($|z| > 3$) of held-out targets, by regulatory distance, for the real network and for controls with permuted edge signs or rewired edges (Wilson 95% intervals, pooled over draws); the controls exceed 0.5 because most responding genes move in the same direction after a knockdown. **(c)** A mechanism report for one held-out K562 target: predicted sign (marker) against observed response (fill), supporting databases below. **(d)** Real-network accuracy at the first hop versus later hops, and the correlation between predicted and observed responses across reached genes.

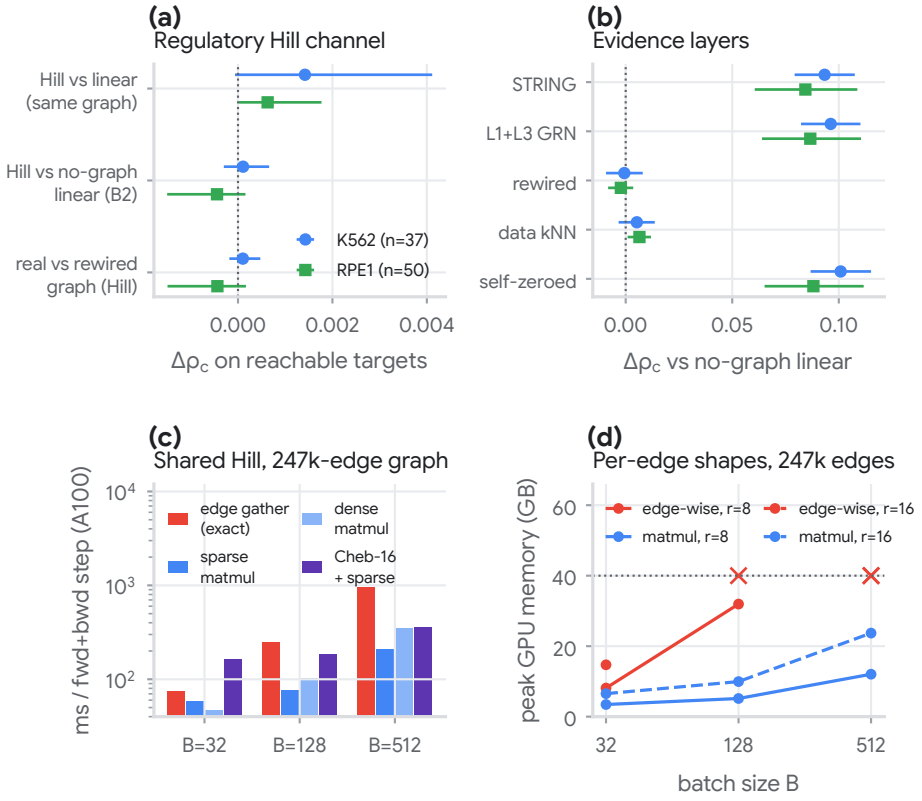


Fig. 8. Ablations and efficiency. **(a)** Hill-form versus linear propagation on the curated signed core, reachable test targets only (differences in ρ_c; 37 K562 and 50 RPE1 unique targets). **(b)** Change in ρ_c relative to the no-graph linear model for evidence layers and controls (STRING only; L1+L3; rewired; graph learned from training data; L1+L3 with own-target entries zeroed). **(c)** Time per forward and backward step for the regulator-shared Hill operator on the 247k-edge association graph (A100). **(d)** Peak memory for edge-specific Chebyshev shapes (crosses: out of memory on 40 GB).

Table 6. Convergence certificates on the curated signed graphs (all values generated from the run). Linear operator: $\rho(|W|)$ before scaling, scaled value, and Neumann terms needed for 10^{-6} (median/max over 40 clamped genes); q_v and the largest ratio of observed truncation error to the bound of Theorem 2(iii) (≤ 1 confirms the bound). Certified Hill-form operator at extreme and default parameters: contraction constants, steps to 10^{-6} from four random starts, spread of the solutions and fixed-point residual.

	K562	RPE1
$\rho(\|W\|)$ raw $\rightarrow$ scaled	$8.5 \rightarrow 0.90$	$9.3 \rightarrow 0.90$
Neumann terms to 10^{-6} (median / max)	12 / 20	11 / 17
q_v; truncation-bound ratio	0.977; 0.019	0.983; 0.014
q (max over parameters); q_η	0.500; 0.750	0.500; 0.750
Steps to 10^{-6} (max)	36	44
Spread of solutions; residual	2.2×10^{-6}; 6.6×10^{-7}	3.0×10^{-6}; 5.4×10^{-7}